



Achieving Firm Performance through Supplier Innovation and Service Stakeholder Commitment

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ABSTRACT

A dry port is an inland region or intermodal facility that is linked directly to a seaport. Cikarang Dry Port, while being one of the best-performing dry ports in Indonesia, contributed just 18% of Tanjung Priok Port's loading and unloading volume. This research aims to determine the effect of supplier innovation and service stakeholder commitment on the performance of dry port businesses. The data was gathered through a questionnaire elicited 55 answers from Cikarang Dry port and marine logistics firms in Greater Jakarta. Multiple regression was used to test hypotheses. This research shows that Supplier Innovation and Stakeholder Commitment to Service have a beneficial effect on port company performance. This research inspires managers to identify the beneficial outcomes of Supplier Innovation amongst service stakeholder engagement organizations to enhance port performance throughout the port supply chain.

CCS CONCEPTS

• **Social and professional topics** → Professional topics; Management of computing and information systems; Implementation management; Computing equipment management; • **Applied computing** → Operations research; Industry and manufacturing; Supply chain management;

KEYWORDS

Cikarang, Dry Port Firm Performance, Service Stakeholder Commitment, Supplier Innovation

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1 INTRODUCTION

According to the 2021 World Competitiveness Yearbook (WCY) Survey conducted by the Institute of Management Development (IMD), Indonesia's competitiveness is ranked 37th out of 64 registered countries. Indonesia's ranking in 2021 has risen slightly from its 2019 position of 40 [1]. One of the factors affecting Indonesia's worldwide competitiveness in logistics is the issue of logistics costs. Thus, one of the government's objectives for reducing national logistics costs is to decrease dwell time, defined as the time when containers of imported products are unloaded from transportation facilities and when the commodities depart the port. Indonesia's residency period is still about 3-4 days, making it the slowest ASEAN [2]. One of the government's initiatives to reduce stay time is optimizing dry ports, which are land container terminals that serve as "extended gates" for seaports, allowing for the issuance of imported containers. It occurs directly at the dry dock, without previous customs clearance at the port, which speeds up and simplifies the delivery procedure. The government anticipates that the dry port's functioning will reduce up to two (two) days in residency time. This situation, however, has not manifested as anticipated owing to a variety of reasons. Cikarang dry port, being one of the top-performing dry ports in Indonesia, contributed just 18% to the loading and unloading volume at Tanjung Priok port, falling short of the original estimate of 60%. [3] The performance of land ports is influenced by factors such as improved road access to land and rail terminals, which enables increased shipping volume and cargo traffic growth [4].

The pandemic nature of Covid-19 necessitates that providers innovate in their product or service development. The capacity of suppliers to create a new processing market for a new product is referred to as supplier innovation [5]. Organizations need their supply chain partners to receive innovations. Additionally, supply chain companies seek to spur innovation outside and minimize investment. Suppliers play a critical role in firm performance, and they are increasingly responsible for pushing innovation in techniques, design, and processes. Buyers in the supply chain are attempting to incentivize suppliers to embrace innovation. Nonetheless, supplier innovation confronts many obstacles, including consumer capability, social and topographical differences, growing company risks, and direct expenses. However, supplier innovation and its role in generating varied firm performance have received little attention so far [6]

Stakeholders (interested parties) are organizations, groups, and people that can affect or impact a business's operations [7]. Numerous parties are engaged in or interested in optimizing the dry port function in Indonesia, including terminal operators, shipping lines, railway operators, trucking firms, logistics service providers, customers (shipper/consignee), and storage businesses. Customs and administrative institutions [3]. Thus, each stakeholder engaged in improving a firm's success must demonstrate a high level of dedication. Although shipping continues to operate despite pandemic-related delays, the issue highlights the performance difficulties faced by ports, which are already extremely susceptible to disorders caused by climate change and natural weather disasters. According to the explanation above, the purpose of this research is to ascertain the effect of supplier innovation and stakeholder commitment on firm performance in the dry port industry.

2 LITERATURE REVIEW

2.1 Firm Performance

A dry port (or land port) is an inland region or intermodal facility linked directly to a seaport. Carriers may now pick up or drop off goods at the intermodal terminal from this location. The port's dry dock provides the same level of service as the port, including container storage, maintenance, cleaning, and customs clearance. This dry port may be advantageous for developing vital regions around the harbour [8]. The World Bank developed a standard for measuring port performance, namely the Logistics Performance Index (LPI). The LPI of a port is a macroeconomic indicator used to help combine data into a standard needle. The LPI is evaluated based on six main criteria: (1) efficiency of the permitting process (2) quality of infrastructure related to trade and transportation; (3) ease of organizing shipments at competitive prices; (4) competence and quality of logistics services; (5) the ability to track and trace shipments; (6) the timeliness of delivery to reach the destination by the scheduled or expected delivery time [9].

2.2 Supplier Innovation and Firm Performance

Suppliers are willing to invest capital in improving innovation, operational efficiency, and high quality of products and services to earn the trust of their customers. Supply innovation can protect the logistics business from various risks, with innovative alternatives and redesigning existing products or services. Supplier innovation plays an important role in reducing costs and improving service quality [10]. Networking between companies promotes the flow of information and the collection and exchange of new ideas. Port business providers can search, process, absorb and share information and experience to generate greater synergies than the information processing capabilities of a company with a single provider [11].

Suppliers with different cultures and areas of expertise provide fertile ground for innovation. That the more suppliers, the greater the supplier's innovative capacity. When the complexity of the supply base reaches a point, that is, there are too many independent providers in a complex system, the activities of the providers tend to become random and unstructured [12].

On the condition that the port company gradually reduces many suppliers, transaction costs and supply risk decrease, while supplier response and innovation increase, leading to higher organizational

performance. However, after the number of suppliers is reduced, the supply risk increases while supplier innovation decreases with a continuous decrease in suppliers, resulting in a decrease in the company's performance [13]. The participation of the suppliers is considered a determining factor of the internal capacities and the company's performance. The importance of supplier innovation is highly appreciated in new companies and large companies [14].

Based on the previous arguments, it can be stated that Hypothesis 1 (H1) There is a positive and significant relationship between Supplier Innovation and the Firm Performance

2.3 Service Stakeholder Commitment and Firm Performance

Stakeholders (interested parties) are organizations, groups, and people that can affect or impact a business's operations [7]. Stakeholders have personal stakes in the business's success and existence. Stakeholders are classified according to whether they are within or external to the organization. Shareholders, employees (including executives, managers, and labourers), and board members are all considered internal stakeholders. Customers, suppliers, alliance partners, creditors, unions, communities, different levels of government, and the media are all examples of external stakeholders. According to Hitt, Ireland, and Hoskisson (2017), Capital markets, market goods, and organizational stakeholders may all be classified as stakeholders. Each stakeholder makes a unique contribution to the business, which results in various advantages for each stakeholder. Employees provide their time and abilities to the organization in exchange for compensation and benefits. Shareholders invest their money in the firm, hoping that the stock will appreciate and the company will pay dividends. Communities are responsible for providing housing, infrastructure, and public safety. They want the corporation to pay taxes, create employment, and refrain from polluting the environment in exchange. As a result, businesses are entangled in a complex exchange relationship with various internal and external stakeholders [15]. Effective stakeholder management may contribute to the success of a business. [7].

Stakeholders are more cooperative and hence more inclined to provide information, which increases the company's value generation or cost reduction; Increased trust reduces a business's transaction expenses; Effective stakeholder network management may result in increased organizational agility and flexibility; Negative outcomes may be minimized, resulting in more predictable and reliable returns; Businesses may establish a strong reputation in the marketplace that is valued by business partners, workers, and consumers. Various stakeholders (stakeholders) in supply chain management must commit to three key features, including service (level of service), cost (cost), and cash (working capital). In terms of service (quality of service), stakeholders must agree to how the partnership will provide the level of service demanded by consumers, including extensive facilities and services and operational flexibility. Meanwhile, stakeholders must commit to the cost savings that their trading partners can afford, including operational expenditures, research and development, sales, and depreciation or amortization. Additionally, supply chain stakeholders must commit to the partners' financial capacity (working capital) to satisfy any payment commitments resulting from the partnership.

Table 1: Statistical Variables of Supplier innovation (X1): Processing results with SPSS 25.0

	Valid	55
	Missing	0
Mean		35.69
Std. Deviation		4.855
Minimum		38
Maximum		56
Sum		3144

In the context of the dry port business supply chain, a service engagement is an agreement between a stakeholder and the dry port company to deliver or utilize facilities or services. Cost commitment refers to the parties' commitment to rationalize the dry ports company's operational expenses to provide competitive rates to its clients. Meanwhile, the cash commitment represents the stakeholder's commitment to the dry port company's cash flow optimization to fund its operations. Port operators, railway operators, customs, land transport businesses, logistics or freight forwarding companies, and storage organizations are all stakeholders in dry port logistics operations [16] [17] [18].

Based on the arguments raised, it can be stated that Hypothesis 2 (H2) There is a positive and significant relationship between the Service Stakeholder Commitment and Firm Performance.

3 RESEARCH METHODOLOGY

This research included a stratified random sample procedure and questionnaires with Likert scale items ranging from 1 to 5. The questionnaire was disseminated through email to 80 workers using a database built with the assistance of businesses and dry port operators to facilitate the online distribution of questionnaires to collect survey data. The purpose of this research is to guarantee that all participants have a minimum of one year of job experience. The email includes a cover letter, a summary of the survey's goals, and a link to the survey's online component. As a consequence, about 55 correct responses were obtained. The 18 elements to measure the firm performance that consists of 6 dimensions adopted from Rezaei [9] : (1) efficiency of the permitting process (2) quality of infrastructure related to trade and transportation; (3) ease of organizing shipments at competitive prices; (4) competence and quality of logistics services; (5) the ability to track and trace shipments; (6) the timeliness of delivery to reach the destination by the scheduled or expected delivery time. The eight items to measure the supplier innovation were adopted from Kim M, Chai S [19] and the nine items to measure the Service Stakeholder Commitment are adopted from Lalam, E. Y. [20]

3.1 Data analysis

3.1.1 Supplier innovation (X1). Based on the results of data collection through questionnaires, the variable Supplier innovation (X1) as one of the factors that affect Firm Performance has the following statistical processing

Table 2: Statistical Variables of Service Stakeholder Commitment (X2): Processing results with SPSS 25.0

	Valid	55
	Missing	0
Mean		33.85
Std. Deviation		5.889
Minimum		33
Maximum		59
Sum		3144

Table 3: Statistical Variables of Firm Performance (Y): Processing results with SPSS 25.0

N	Valid	55
	Missing	0
Mean		47.28
Std. Deviation		8.264
Minimum		38
Maximum		62
Sum		3754

3.1.2 Service Stakeholder Commitment (X2). Based on the results of data collection through questionnaires, the variable Service Stakeholder Commitment (X2) as one of the factors that affect Firm Performance has the following statistical processing:

3.1.3 Firm Performance (Y). Based on the results of data collection through questionnaires, the variable Firm Performance (Y), which is the dependent variable, has the following statistical processing:

4 RESULT

In this study, multiple regression analysis was used to test the hypothesis. According to table 4, showing Supplier innovation, Service Stakeholder Commitment has a significant effect on Firm Performance. The value of F is 20.887. To determine the impact of Supplier innovation and Service Stakeholder Commitment simultaneously on Firm Performance, the results obtained are 35.6% and 18.4%, 24.7% for different effects. This study recommends that all Supplier innovation and Service Stakeholder Commitment variables support the Firm Performance variable.

5 CONCLUSION

Innovation is a typical process that enables enterprises to grasp new markets, remove stagnation and decline, and challenge current businesses even in established industries. Therefore, the world around us, and therefore management techniques, are changing and flourishing in a changing environment management methods must also change. In firms that profit from supplier innovation and information exchange between the supplier and other supply chain components, long-term relationships with supply chain partners are promoted via social norms and trust rather than through legally binding and rigid contracts. Sharing will be critical in such businesses to establish coordination and collaboration with supply

Table 4: Anova Multiple Regression Table (ANOVA(b)): a Predictors: (Constant), Supplier innovation, Service Stakeholder Commitment, b Dependent Variable: Firm Performance

	Sum of Squares	df	Mean Square	F	Sig.
Regression	747.185	2	128.447	20.887	.000(a)
Residual	1.857.564	53	1.745		
Total	1.951.865	55			

chain partners. Supplier innovation is concerned with the supply of initial resources; as a result, the purchase and procurement strategy is critical, and instances timely raw material delivery, minimizing the backwardness caused by rapid technological changes, ensuring adequate material transportation, maintaining adequate inventory and cost savings are all characteristics of supplier innovation.

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